

Hitesh J Gowda

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EDUCATION

Dayananda Sagar College of Engineering <i>B.E. in Electronics and Communication Engineering</i> CGPA (3rd semester): 8.6/10	Bengaluru, Karnataka <i>Expected May 2028</i>
Deeksha PU College <i>Pre-University Course (PCMC)</i> Percentage: 95%	Bengaluru, Karnataka <i>2022 – 2024</i>
RNS Vidyanyikethan <i>Secondary School (Science)</i> Percentage: 97%	Bengaluru, Karnataka <i>Graduated 2022</i>

PROJECTS

AI PDF Chat Assistant <i>FastAPI, Express.js, LangChain, ChromaDB, all-MiniLM-L6-v2, Gemini API</i> - Developed a full-stack AI application using Retrieval-Augmented Generation (RAG) to enable conversational question answering over uploaded PDF documents. - Implemented semantic retrieval using the all-MiniLM-L6-v2 embedding model and ChromaDB for persistent vector storage, eliminating embedding API costs. - Integrated history-aware retrieval to support contextual follow-up questions and improve response relevance. - Designed a modular architecture with Express.js handling file uploads and API routing while FastAPI served the AI pipeline.	GitHub
RAG Pipeline from Scratch <i>Python, FastAPI, NumPy, all-MiniLM-L6-v2, Gemini API</i> - Implemented every stage of the Retrieval-Augmented Generation (RAG) pipeline from scratch without using LangChain, LlamaIndex, or vector databases. - Built custom modules for document loading, chunking with overlap, embedding generation, cosine similarity retrieval, prompt construction, and context grounding. - Developed semantic search using NumPy-based cosine similarity and integrated Gemini API for context-aware answer generation. - Designed a modular FastAPI application following clean architecture principles with documented REST APIs.	GitHub
Video Sharing Platform Backend <i>Node.js, Express.js, MongoDB, JWT, Cloudinary</i> - Developed a RESTful backend supporting user authentication, video uploads, comments, playlists, subscriptions, likes, tweets, and watch history. - Implemented JWT authentication, refresh tokens, secure route protection, and ownership-based authorization across application resources. - Integrated Cloudinary and Multer for scalable media upload and storage workflows. - Designed modular APIs with pagination, search functionality, and MongoDB aggregation for efficient data retrieval.	GitHub

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++
Frameworks: FastAPI, Express.js, Node.js, LangChain, SQLAlchemy, Mongoose
Databases: PostgreSQL, MongoDB
Libraries & APIs: NumPy, Sentence Transformers, Gemini API, Groq API, OpenAI API
Developer Tools: Git, GitHub, Linux, Postman, VS Code